

VARUN KIRAN JOSHI

Mumbai, Maharashtra, India • +91 7715881457 • varunjoshi311004@gmail.com

linkedin.com/in/varunjoshi3110 • github.com/Varun311004 • Portfolio (In Development)

PROFESSIONAL SUMMARY

Computer Engineering graduate with practical experience developing production-ready full-stack, mobile and AI-powered applications through industry internship and collaborative engineering projects. Contributed to AI-powered software, IoT applications, secure authentication systems, and scalable backend integrations while working with cross-functional teams. Strong foundation in software engineering principles, problem-solving, and building reliable applications, with a growing interest in full-stack development, cloud technologies, and applied artificial intelligence.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering

New Horizon Institute of Technology and Management, Thane, Maharashtra

CGPA: 7.39/10 | September 2023 – June 2026

Diploma in Computer Engineering

Vidya Prasarak Mandal's Polytechnic, Thane, Maharashtra

Percentage: 80.06% | June 2020 – June 2023

Secondary School Certificate (SSC)

Sister Nivedita English School, Dombivli, Maharashtra

Percentage: 88.60% | March 2020

TECHNICAL SKILLS

Programming Languages: Dart, Java, Python, JavaScript, PHP, SQL, HTML5, CSS3

Frontend Development: Flutter, React, Bootstrap, Responsive Web Design

Backend Development: Node.js, REST APIs, JWT Authentication

Artificial Intelligence: LangChain, Retrieval-Augmented Generation (RAG), Whisper Tiny, TinyLlama, Sentence Transformers (MiniLM), TensorFlow.js

Databases & Storage: MongoDB, PostgreSQL, Firebase Realtime Database, Cloudflare R2, phpMyAdmin

Developer Tools: Git, GitHub, VS Code, Android Studio, Postman, Firebase, Vercel, Render

Core Concepts: Object-Oriented Programming (OOP), State Management (BLoC, Provider), Bluetooth Low Energy (BLE), API Integration, Software Development Life Cycle (SDLC)

PROFESSIONAL EXPERIENCE

FLUTTER DEVELOPER INTERN

Manous Innovation Pvt. Ltd. | Pune, Maharashtra

April 2025 – September 2025

- Developed and maintained 10-12 production Flutter screens for an IoT-based smart mobility application, implementing user onboarding, authentication, role-based access, Bluetooth Low Energy (BLE) connectivity, and user workflows.
- Integrated REST APIs using structured state management (BLoC and Provider) to support secure authentication and seamless communication between the mobile application and backend services.
- Collaborated with backend engineers working with Java Spring Boot and PostgreSQL to integrate APIs, troubleshoot issues, and deliver production-ready features.
- Developed a fully functional authentication module as part of the company's technical hiring assessment; the implementation was later incorporated into the production application.

- Used Git/GitHub and Postman for collaborative development, API testing, debugging, version control, and integration within an Agile engineering environment.

WEB DEVELOPER INTERN

LazyTech InfoTech Solutions | Dombivli, Maharashtra

July 2022 – August 2022

- Developed responsive user interfaces for e-commerce websites using HTML, CSS, JavaScript, and Bootstrap.
- Assisted in integrating MySQL databases to support dynamic content and backend functionality.
- Collaborated with the development team to debug issues, improve website responsiveness, and contribute to a live client project while following software development best practices.

PROJECTS

Cortexa – AI Powered Educational Platform

Duration: August 2025 – April 2026

Role: Flutter Developer & AI Integration Contributor (3-member team)

Tech Stack: Flutter, React, Node.js, Python, MongoDB, Cloudflare R2, LangChain, TinyLlama, Whisper Tiny, Sentence Transformers (MiniLM), JWT Authentication, BLoC, Vercel

Overview: Designed and developed a multi-tenant educational platform that enables institutes to securely manage learning resources while providing students with an AI-powered assistant capable of answering questions using Retrieval-Augmented Generation (RAG), generating MCQs, and performing speech-to-text transcription.

Responsibilities:

- Led the complete development of the Flutter application, independently designing and implementing 20+ responsive screens using the BLoC architecture, while integrating backend APIs and AI-powered features.
- Contributed to the implementation of the Retrieval-Augmented Generation (RAG) pipeline by integrating PDF upload workflows, document parsing, chunk retrieval, embeddings generated using Sentence Transformers (MiniLM), and AI-generated responses using TinyLlama.
- Developed and integrated 2-3 backend APIs supporting teacher and student workflows while collaborating with teammates on frontend enhancements, backend functionality, and AI feature development.
- Implemented secure authentication using JWT-based authorization for institute-specific and user-specific access across the platform.
- Integrated Whisper Tiny for speech-to-text functionality and collaborated on AI-assisted MCQ generation to enhance the digital learning experience.
- Worked with Cloudflare R2 for document storage, MongoDB for metadata and document management, and supported deployment of the web platform on Vercel.

Elevate – AI-Powered Adaptive STEM Learning Platform

Duration: November 2025 – April 2026

Role: Full-Stack & AI Developer (Independent Development for 3-member team)

Tech Stack: Python, Flask, Flask-SQLAlchemy, Alembic, PostgreSQL, SQLite, FAISS, ChromaDB, Scikit-learn, MediaPipe, TensorFlow.js, HTML5, CSS3, JavaScript

Overview: Designed and developed an AI-powered adaptive learning platform that personalizes STEM education by combining emotion recognition, learning analytics, adaptive content delivery, and AI-assisted tutoring to improve student engagement and learning outcomes.

Responsibilities:

- Contributed to the development of a full-stack adaptive learning platform by integrating AI services, backend APIs, and modern web application features while collaborating within a multidisciplinary development team.
- Developed responsive frontend modules using HTML5, CSS3, and vanilla JavaScript while integrating REST APIs to support secure JWT authentication, personalized learning workflows, and seamless client-server communication.
- Worked on AI-driven adaptive learning features by contributing to emotion-aware learning recommendations, Bayesian Knowledge Tracing (BKT), and intelligent RAG-based tutoring workflows.

- Assisted in integrating browser-based computer vision (MediaPipe/TensorFlow.js) for real-time emotion recognition to enhance learner engagement and adapt educational content dynamically.
- Collaborated on backend development using Python, Flask and PostgreSQL, supporting secure data management, scalable API communication, and seamless LLM service integration.
- Participated in project architecture discussions, feature planning, debugging, testing, and version control using Git/GitHub throughout the software development lifecycle.

FindEasy – Local Service Provider Mobile Application

Duration: February 2025 – April 2025

Role: Solo Developer

Tech Stack: Flutter, Firebase Authentication, Firebase Realtime Database, Firebase Cloud Messaging (FCM), Google Maps API, BLoC, REST APIs

Overview: Designed and developed a cross-platform mobile application that connects customers with verified local household service providers through location-based discovery, secure authentication, booking management, and real-time communication.

Responsibilities:

- Independently designed and developed the complete application, implementing 15+ Flutter screens for separate customer and service provider workflows with responsive and user-friendly interfaces.
- Built secure user authentication, provider onboarding, booking management, review system, and real-time data synchronization using Firebase Authentication and Firebase Realtime Database.
- Integrated Google Maps API to enable location-based provider discovery and improve user accessibility to nearby services.
- Implemented BLoC state management to separate business logic, improve maintainability, and efficiently manage application state.
- Designed the Firebase database architecture with collections for Users, Providers, Bookings, and Offers, including nested collections for provider reviews and services.
- Integrated Firebase Cloud Messaging (FCM) to support real-time notifications for booking updates and service interactions.

DigiResult – Student Result Management System

Duration: August 2024 – October 2024

Role: Solo Developer

Tech Stack: PHP, HTML5, CSS3, JavaScript, MySQL, SMTP, XAMPP

Overview: Designed and developed a full-stack web application that streamlines student result management through secure authentication, centralized result access, and OTP-based email verification for educational institutions.

Responsibilities:

- Independently designed and developed both the frontend and backend of the application using PHP, HTML5, CSS3, JavaScript, and MySQL.
- Designed the complete relational database schema and implemented secure data management for student records, result processing, and administrative operations.
- Implemented an OTP-based email authentication system using SMTP to verify student identity before granting access to examination results.
- Developed separate administrative functionalities for secure result management while providing students with a centralized portal to access academic results efficiently.
- Configured and tested the application in a XAMPP development environment while following secure coding practices and structured backend development principles.
- Reduced manual result distribution by providing an online platform that simplified access to academic records for students and administrations.

GeoLogr – Geofence-Based Staff Attendance System

Duration: December 2022 – April 2023

Role: Android Developer & Geofencing Contributor (4-member team)

Tech Stack: Android Studio, Java, XML, Firebase Authentication, Firebase Realtime Database, Cloud Firestore, Firebase Storage, Google Maps SDK, Google Geofencing API, Google ML Kit, TensorFlow Lite, OpenCV

Overview: Designed and developed an Android-based staff attendance management system that automates attendance using location-aware geofencing and facial recognition. The application enables administrators to define secure workplace boundaries while ensuring employees can mark attendance only after successful location validation and biometric verification, reducing attendance fraud and improving record accuracy.

Responsibilities:

- Contributed to the Android applications for both administrators and employees using Java and Android Studio, implementing secure attendance workflows and user management features.
- Designed and implemented the geofencing module, allowing administrators to create, update, and manage workplace boundaries by configuring radii using Google Maps SDK and Google Geofencing API.
- Developed location validation workflows that continuously monitored employee location and enabled check-in/check-out actions only after successful entry into the assigned geofenced area, preventing location-based attendance fraud.
- Integrated facial recognition using OpenCV for image preprocessing and histogram equalization, providing secure biometric authentication before attendance registration.
- Worked with Firebase Authentication, Cloud Firestore, Firebase Realtime Database, and Firebase Storage to manage authentication, employee profiles, geofence configurations, attendance records, and facial embeddings.
- Implemented administrative features for employee management, geofence configuration, attendance monitoring, and export of attendance reports while collaborating with teammates through development, testing and project integration.

ACHIEVEMENTS & AWARDS

- Winner – Vibe-a-Thon Coding Competition, developed a React.js-based user interface as part of a competitive web development challenge.
- 5th Place (Consolation Prize) – Smart India Hackathon (Internal College Round) for collaborative problem-solving and software development.
- Participant – UIDAI Hackathon, contributing to solution design and software development.
- Published Research Paper in the International Journal of Research Publication and Reviews (IJRPR) as part of the final-year engineering curriculum.
- Bronze Medallist – State Level Roller Skating (Quads) during school-level sports competitions.
- Blue Belt – Karate, demonstrating discipline and long-term commitment to martial arts.

CERTIFICATIONS

- **The Complete 2024 Web Development Bootcamp** – Udemy (Instructor: Dr. Angela Yu)
- **AWS Cloud Hands-on Workshop** – Computer Society of India (CSI)
- **Google Agentic AI Day** – Google (Badge Earned)
- **AI Artist Workshop** – Google Student Ambassador Program (Participation Certificate)
- **Poster Presentation Competition** – College Level (Participation Certificate)

LEADERSHIP & PROFESSIONAL ACTIVITIES

- **Technical Volunteer**, Training & Placement Cell, assisted in organizing and supporting a campus recruitment job fair by managing technical operations and candidate coordination.

- **Member**, Computer Society of India (CSI), actively participated in technical workshops and industry-oriented events.
- **Member**, Bureau of Indian Standards (BIS) Student Chapter, engaged in technical awareness programs and standardization initiatives.

HACKATHONS & TECHNICAL EVENTS

- **Google Agentic AI Day Hackathon** (Online) – Participated in a team-based hackathon focused on building AI agent applications using Google's Agentic AI ecosystem.
- **UIDAI Hackathon** – Participated in developing technology-driven solutions for digital identity and public service challenges.
- **Smart India Hackathon (Internal College Round)** – Secured 5th place while developing an innovative software solution as part of a multidisciplinary team.

PUBLICATIONS

CortexaX: An AI Powered Educational Platform

International Journal of Research Publication and Reviews (IJRPR) | Volume 7, Issue 4, April 2026

- Published research describing the design and implementation of an AI-powered educational platform leveraging Retrieval-Augmented Generation (RAG), semantic search and intelligent tutoring to enhance personalized learning experiences.
- **Paper ID:** IJRPR20262528
- **DOI:** 10.55248/gengpi.07.0426.20835

PROFESSIONAL INTERESTS

- **Artificial Intelligence**
- **Full-Stack Software Development**
- **Mobile Application Development (Flutter)**
- **Cloud Computing**
- **Financial Technology (FinTech)**

LANGUAGES

- **English** — Professional Working Proficiency
- **Hindi** — Native Proficiency
- **Marathi** — Native Proficiency